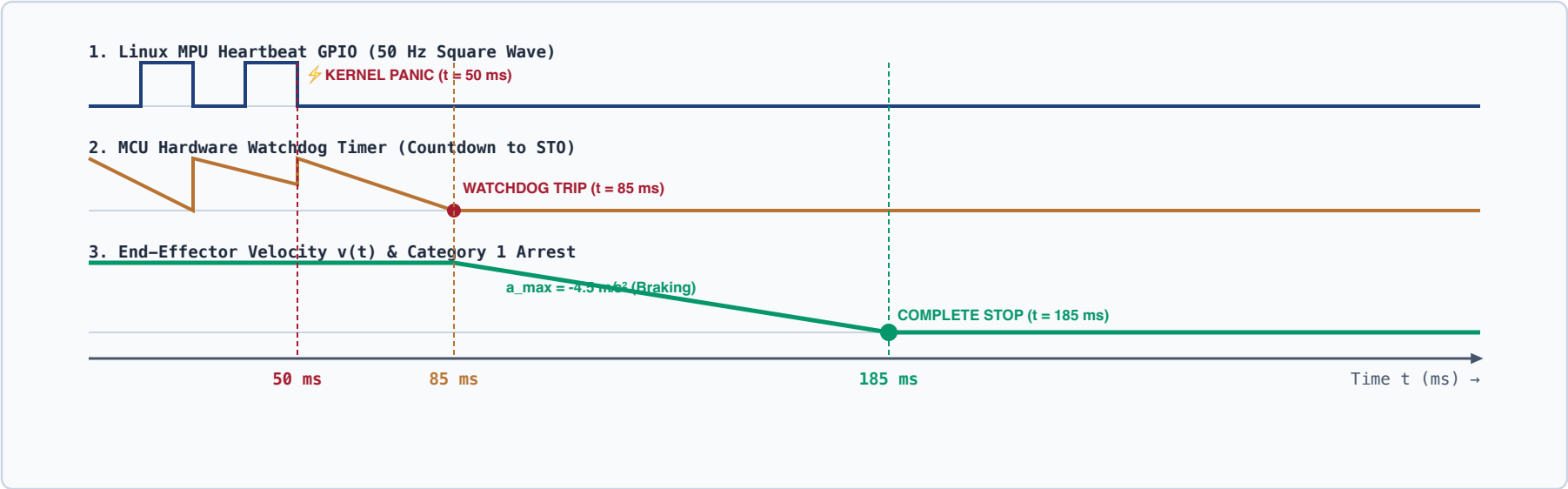


CROSS-LAYER SEEDED FAULT INJECTION & DYNAMIC ARREST WAVEFORM

Live telemetry: Linux MPU kernel crash at $t = 50\text{ ms}$ → Watchdog trip at $t = 85\text{ ms}$ → Safe arrest at $t = 185\text{ ms}$ ($d_{\text{stop}} = 5.2\text{ cm} < 12.0\text{ cm}$)



✓ PHYSICAL VERIFICATION: BOUNDED STOPPING MARGIN PROOF

Measured Stopping Distance:

$d_{\text{stop}} = 5.2\text{ cm}$ (Reaction: 1.6 cm + Braking: 3.6 cm)

Obstacle Clearance Envelope:

$d_{\text{clearance}} = 12.0\text{ cm}$

VERIFIED SAFETY MARGIN:

+6.8 cm (Zero Contact!)

Conclusion: Even under total MPU hardware failure, the Reflex Brain brings the robot to a complete rest within physical clearance envelopes.